OpsArena: Production Cloud-Native Incident Response RL Environment

TECHNICAL DESIGN DOCUMENT & VERIFIER SPECIFICATION | TEAM_087

Team Code: team_087	**Track:** Software & DevOps / Business Ops	**Evaluated Model:** gpt-6-luna (penpencil.guru)	**Pre-Flight Baselines:** **100% Passed (8/8 in 2.77s)**

1. Executive Summary & Problem Formulation

Modern cloud operations require autonomous AI agents to diagnose, triage, and remediate cascading incidents across distributed microservices under strict Service Level Objectives (SLOs: error rate < 0.1%, p99 latency < 100ms) and operational budgets ($150.00). Existing RL benchmarks either test synthetic toy math problems or supply unconstrained Docker sandboxes vulnerable to prompt injection, sandbox escapes, and reward tampering. OpsArena bridges this gap with a deterministic, hardened, discrete-event simulation of CloudMesh (8 microservices), a dual-container isolated evaluation harness, an 8-task procedurally seeded benchmark suite, and a mathematically gated continuous multi-factor verifier.

2. Dual-Container Architecture & Sandboxing Threat Model

• **Network Segmentation:** The harness container connects to the external model API (internet) and an internal bridge (isolated). The sandbox container runs exclusively on the isolated network (internal: true) with zero public internet egress.
• **Zero Host Socket Access:** The Docker daemon socket (/var/run/docker.sock) is strictly never mounted. Process breakout and sibling container manipulation are eliminated by design.
• **Least Privilege:** The sandbox runs as an unprivileged user (UID 1000: sandbox), enforces cap_drop: ALL, no-new-privileges, a 512MB RAM ceiling, and a 256 PIDs limit.
• **Verifier Inaccessibility:** All evaluation criteria, ground truth thresholds, and reward calculation logic reside solely inside the harness. The sandbox exposes only /health, /reset, /tool, and /state, preventing reward tampering.
• **Input Sanitization & Output Bounds:** All operational tools enforce key-specific schemas and service scopes (e.g. setting database pool size on api-gateway returns a clean error). Tool outputs strictly clip at 4,000 characters including truncation markers, and in-flight logs regex-scrub passwords, API keys, and bearer tokens with [REDACTED].

3. Operational Action Space (8 Schema-Validated Tools)

Tool	Cost	Parameters & Type Bounds	Operational Function & Safeguards
`get_topology`	$1.00	None	Inspects 8 services: pod replicas, versions, dependencies, and cumulative spend.
`query_metrics`	$2.00	service, metric (latency, errors, cpu, mem, qps, rps)	Unified metrics engine: returns 5-step timeseries with status (HEALTHY / CRITICAL). No budget leak on bad inputs.
`query_logs`	$3.00	service, filter (case-insensitive substring)	Structured log query with automated regex secret sanitization and 4,000 char bounds.
`scale_service`	$2/pod+$3	service, replicas (int: 1 to 20)	Scales pod pool within platform boundaries; rejects non-ints and out-of-range values.
`rollback_release`	$20.00	service, target_version (string)	Rolls back faulty deployments to verified stable container images in release registry.
`update_config`	$5.00	service, key, value (service-scoped typed)	Service-specific schema: enforces int bounds (pool: 1-100, lock: 0-60k) and booleans. Rejects unauthorized keys.
`reroute_traffic`	$10.00	service (order-service), weight (int: 0 to 100)	Shifts canary traffic weights (0% to 100%) to safely isolate failing routes.
`finish`	$0.00	None	Signals completion of incident mitigation and locks state for formal verification scoring.

4. Multi-Factor Programmatic Reward Formulation

OpsArena computes a multi-factor continuous reward gated by an absolute incident resolution indicator: **Reward = P × [0.45 × S_SLO + 0.25 × R_reg + 0.15 × B_bud + 0.15 × E_eff]**.
• **Progress Multiplier (P ∈ [0.0, 1.0]):** P = 0.00 for no-op/idle agents, guaranteeing a score of **0.0000** ≤ **0.05**. Bounded partial credit (P ∈ (0, 0.5]) is awarded for incremental progress (e.g. partial scaling), and P = 1.00 upon full incident remediation.
• **SLO Recovery (S_SLO ∈ [0, 1]):** Continuous score evaluating post-incident error rate (≤ 0.1%) and p99 latency (≤ 100ms) with dual-layer telemetry agreement.
• **Platform Stability (R_reg ∈ [0, 1]):** Compares all non-target microservices against pre-incident baseline health. If an agent downscales, rolls back, drains, or misconfigures an unaffected service, R_reg drops sharply, penalizing regressions.
• **Budget Preservation (B_bud ∈ [0, 1]):** Full marks for spend ≤ $50.00, decaying linearly to 0 at the $150.00 operating budget ceiling.
• **Action Efficiency (E_eff ∈ [0, 1]):** Full marks for surgical resolution within ≤ 8 total model turns, decaying linearly up to 40 turns.

5. Procedurally Seeded 8-Task Incident Benchmark Suite

All 8 task instructions state business symptoms, observable alerts, and SLO constraints without leaking hidden root causes or exact tool commands:

Task ID	Scenario Title	Diff	Simulated Root Cause & Seeded Variations	Target SRE Action	Oracle	No-Op
`task_01`	Redis Eviction Storm	Easy	Mem capacity (1-4GB), surge QPS (2.8k-3.6k) causes DB CPU 96%.	Scale cache-cluster ≥ 2 pods	0.996	0.000
`task_02`	Bad Deployment Rollback	Easy	Faulty release tag throws NullPointerExceptions (16-21% 500s).	Rollback order-service to v2.4.0	0.996	0.000
`task_03`	DB Pool Starvation	Med	Checkout bursts (1.6k-2.2k RPS) exhaust pool (504 timeouts).	Update db_pool_size ≥ 25	0.996	0.000
`task_04`	Slow Third-Party Gateway	Med	External payment partner hangs (6s-9s), causing thread starvation.	Enable circuit_breaker_enabled	0.996	0.000
`task_05`	Cascading Failure	Hard	Flash surge (4.4k-5.2k RPS) overwhelms pods, causing OOM crash loops.	Scale order-service ≥ 6 pods	0.994	0.000
`task_06`	Canary Memory Leak	Med	Severe memory leak on canary split route (5-15% traffic).	Reroute traffic weight to 0%	0.996	0.000
`task_07`	Overprovisioned Drain	Hard	Cluster overscaled (36 pods), burning $540/hr under 12% CPU. Safe min enforced.	Right-size cluster to ≤ 18 pods	0.996	0.000
`task_08`	Migration Lock Starvation	Hard	Exclusive table lock blocks writes (22% checkout errors, PID 4k-9k).	Set lock_timeout_ms to 1000	0.996	0.000

6. Empirical Evaluation & Live Autonomous Model Runs

• **Pre-Flight Baseline Gating:** Across all 8 tasks, pre-flight checks completed in **2.77 seconds** (budget: 120s), with 100% pass rate (Oracle Mean: 0.996 ≥ 0.95, No-Op Mean: 0.0000 ≤ 0.05). Invariance verified across multiple seeds (0, 42, 99).
• **Live gpt-6-luna Autonomous Evaluations (penpencil.guru):** Three real model runs were executed and archived in test-run-logs/:
1. **run_01_task_02_bad_deployment:** Diagnosed NPE, rolled back to v2.4.0, verified error rate dropped from 18.4% → 0.04% in 4 turns (12.1s, Reward: **0.9955**).
2. **run_02_task_01_cache_pressure:** Investigated cache eviction storm, scaled cache-cluster to 2 pods, verified CPU normalized in 4 turns (11.9s, Reward: **0.9955**).
3. **run_03_task_05_cascading_failure (Negative Control):** Step cap hit during exploratory reads before issuing scaling command in 1 turn (3.3s, Reward: **0.0000**).
• **Sample Trajectories:** Documented in sample-trajectories/trajectory_success_task_02.md and sample-trajectories/trajectory_failed_task_05.md with full SRE post-mortems.

7. Adversarial Verification & Official Run Contract Compliance

Verification Vector / Contract Requirement	OpsArena Tested Implementation & Defense	Automated Test / Status
Task 7 Undersized Cluster Attack Defeat	Minimum safe capacity enforced per service. Scaling all pods to 1 triggers severe SLO starvation.	`test_task_07_undersized_cluster_attack_defeated [PASS]`
Dynamic Telemetry & Log Invariance	Metrics and logs dynamically reflect seeded bad versions, RPS surges, and partner hangs without drift.	`test_task_02_bad_version_dynamic_telemetry [PASS]`
Credential Scrubbing & 4k Bounds	Regex redacts API keys, bearer tokens, and passwords. Output length strictly clipped at 4k chars with marker.	`test_secret_scrubbing_and_output_bounds [PASS]`
Budget Preservation on Invalid Calls	Invalid service names and unallowable metrics return errors prior to deducting remediation cost.	`test_cost_not_deducted_on_invalid_calls [PASS]`
Non-Target Config Regression Penalties	Mutating configs of unaffected services incurs dynamic regression penalties in R_reg.	`test_config_regression_on_non_target_services [PASS]`
Single-Command Turnkey Contract	docker compose up --build --abort-on-container-exit --exit-code-from harness	`100% Contract Compliant (Exit 0, 1, 2, 3)`